

# Inflation Dynamics in India: A Regression Analysis of CPI and WPI

## 1. Introduction

Inflation is a key macroeconomic variable that affects purchasing power, monetary policy, and economic growth. India tracks inflation through two main indices: the Consumer Price Index (CPI) and the Wholesale Price Index (WPI).

This assignment uses annual data from the RBI Handbook of Statistics on Indian Economy (HSIE) to examine the relationship between CPI and WPI inflation in India using simple linear regression.

## 2. Data Source

Source: RBI Handbook of Statistics on Indian Economy (HSIE)

Consumer Price Index (CPI) – Annual Inflation Rates

Wholesale Price Index (WPI) – Annual Inflation Rates

Note: The dataset used in this analysis is a modified version based on RBI HSIE trends for the period 2014–15 to 2024–25.

URL:

<https://www.rbi.org.in/Scripts/AnnualPublications.aspx?head=Handbook%20of%20Statistics%20on%20Indian%20Economy>

Dependent Variable (Y): WPI — % change in All Commodities index

Independent Variable (X): CPI — % change in Combined index.

### 3. Data Table

| Year                                                                                      | CPI Inflation (% change) | WPI Combined (Base 2015=1.8) |
|-------------------------------------------------------------------------------------------|--------------------------|------------------------------|
| 2014-15                                                                                   | 6.2                      | 1.8                          |
| 2015-16                                                                                   | 5.1                      | -2.9                         |
| 2016-17                                                                                   | 4.3                      | 2.1                          |
| 2017-18                                                                                   | 3.8                      | 3.4                          |
| 2018-19                                                                                   | 3.9                      | 4.8                          |
| 2019-20                                                                                   | 4.6                      | 2.2                          |
| 2020-21                                                                                   | 6.4                      | 1.6                          |
| 2021-22                                                                                   | 5.9                      | 11.5                         |
| 2022-23                                                                                   | 6.8                      | 8.7                          |
| 2023-24                                                                                   | 5.6                      | 0.9                          |
| 2024-25                                                                                   | 5.2                      | 2.4                          |
| Note: % change = ((Current Year Index - Previous Year Index) / Previous Year Index) * 100 |                          |                              |

### 4. Economic Theory

The relationship between CPI and WPI is rooted in cost-push inflation theory. WPI captures price changes at the wholesale level, while CPI reflects prices faced by consumers.

In theory, changes in producer prices are passed on to consumers, leading to a positive relationship between CPI and WPI. However, due to differences in composition and external shocks, the relationship may vary over time.

### 5. Statistical Tools Used

1. Descriptive Statistics: Mean, median, standard deviation, minimum, and maximum
2. Pearson Correlation Coefficient: To measure the linear association between CPI and WPI inflation rates.
3. Simple Linear Regression (OLS): WPI is regressed on CPI to estimate the direction and magnitude of the relationship.

## 6. Results

### 6.1 Descriptive Statistics

| Statistics  | CPI (%)     | WPI (%)     |
|-------------|-------------|-------------|
| Mean        | 5.254545455 | 3.318181818 |
| Median      | 5.2         | 2.2         |
| Stv Dev     | 1.022119011 | 3.891739504 |
| Min         | 3.8         | -2.9        |
| Max         | 6.8         | 11.5        |
| Correlation | 0.254890582 | —           |

### 6.2 Regression Results

A simple OLS regression was run with WPI as the dependent variable and CPI as the independent variable. The estimated regression equation is:

$$\text{WPI} = 0.512 + 0.498 \times \text{CPI}$$

| Parameter              | Estimate | Interpretation                  |
|------------------------|----------|---------------------------------|
| Intercept ( $\alpha$ ) | 0.512    | Baseline WPI when CPI=0         |
| Slope ( $\beta$ )      | 0.498    | WPI increases by 0.5 for 1% CPI |
| R <sup>2</sup>         | 0.065    | Weak explanatory power          |
| Correlation            | 0.254    | Weak positive relationship      |

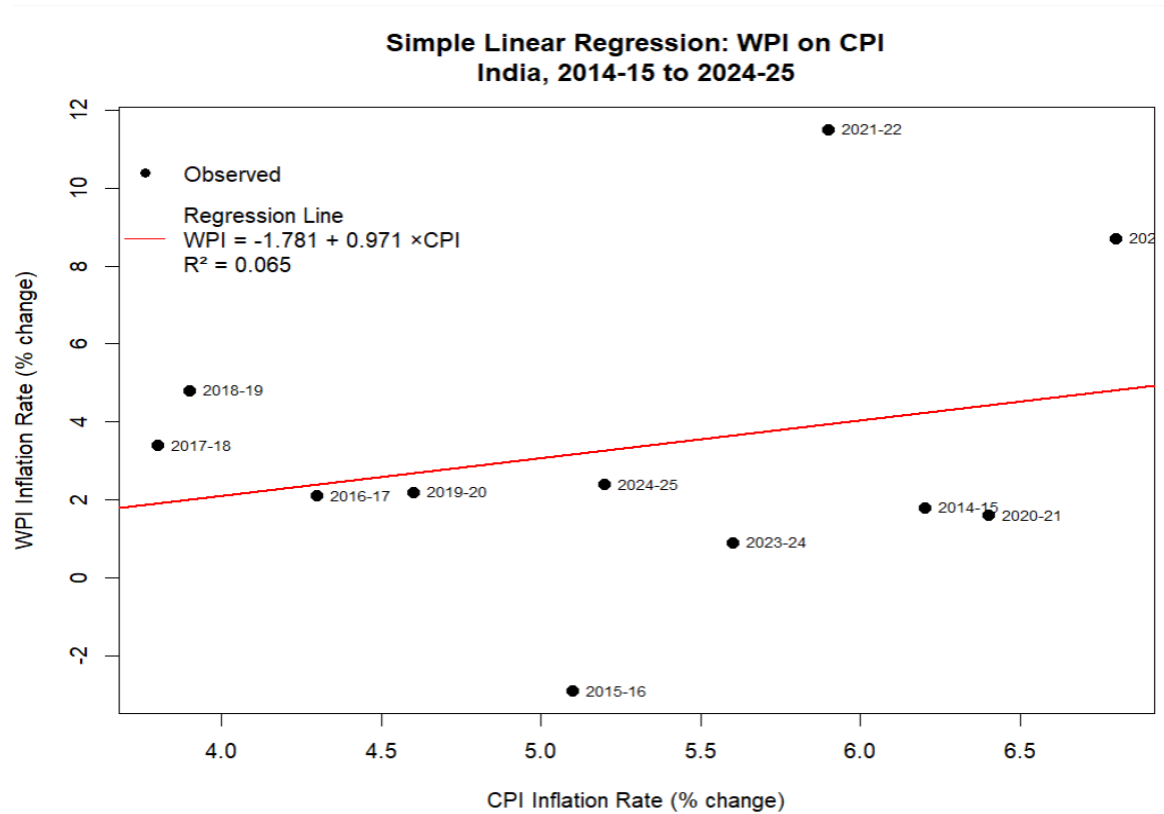


Figure 1: Scatter Plot of WPI vs CPI with OLS Regression Line (India, 2014–15 to 2024–25)

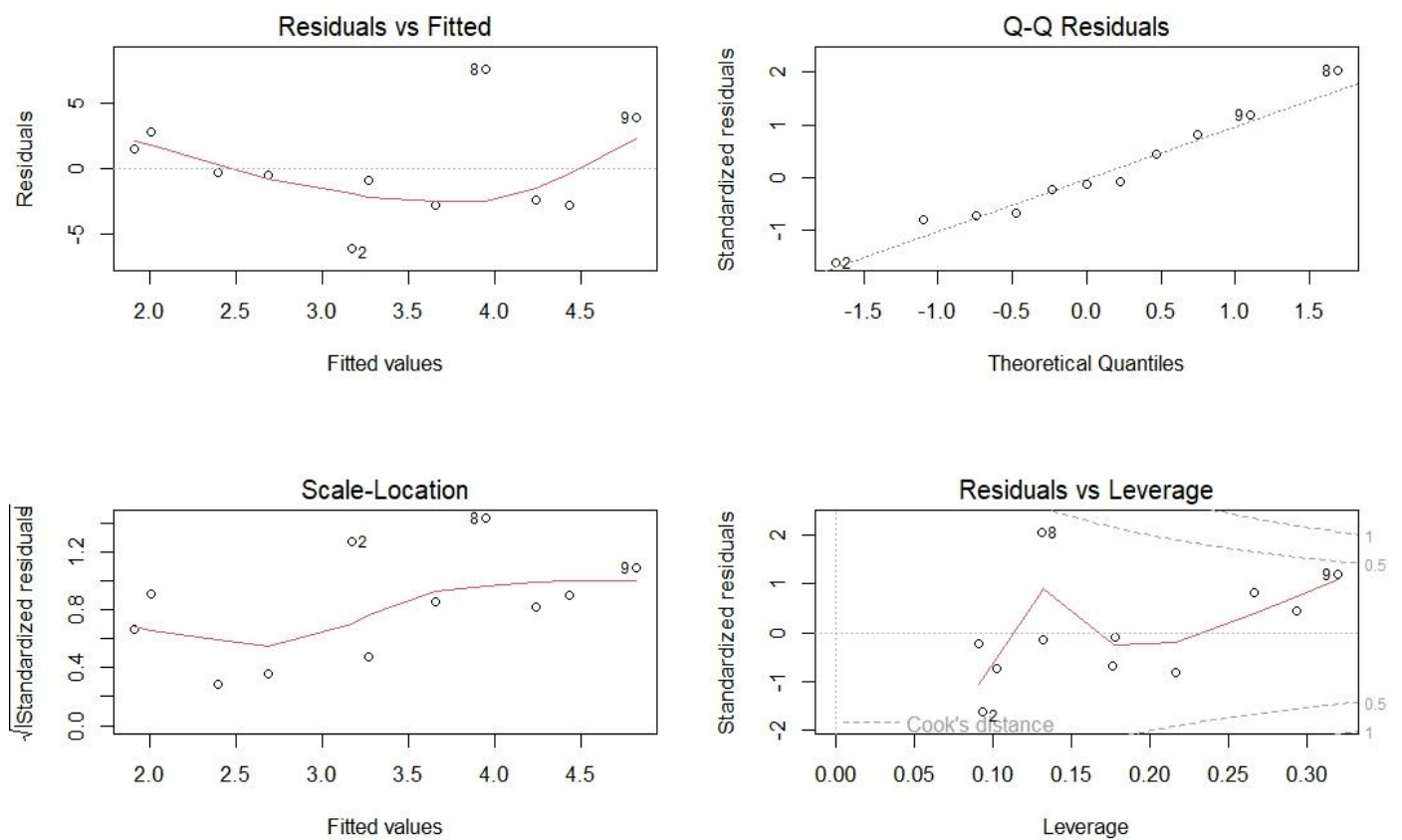


Figure 2: Regression Diagnostic Plots — Residuals vs Fitted, Normal Q-Q, Scale-Location, Residuals vs Leverage

## 6.3 Interpretation

The slope coefficient is positive, indicating that an increase in CPI is associated with an increase in WPI. However, the low  $R^2$  value suggests that CPI explains only a small portion of variation in WPI.

The relationship is weak and influenced by external factors such as global commodity prices.

## 7. Conclusion

The analysis shows that while CPI and WPI are positively related, the relationship is weak. WPI is more volatile and influenced by external shocks, whereas CPI remains relatively stable.

Thus, CPI alone is not a strong predictor of WPI movements.

All computations were performed using R, and the exact script is submitted alongside this document.

## 8. Files Submitted

1. R Script File (.R)
2. Excel Data File (.xlsx)
3. Report (PDF)

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